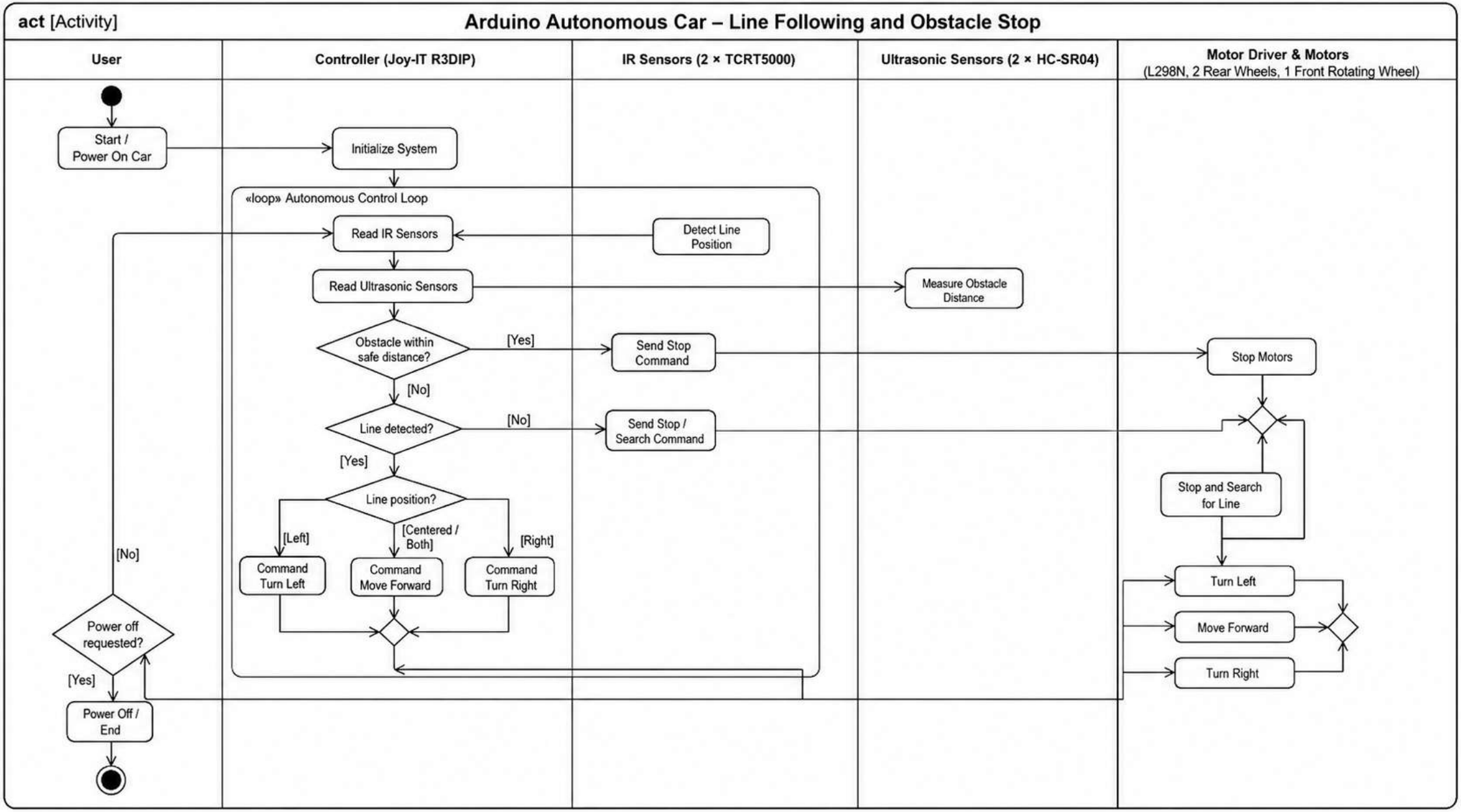

act [Activity]
Arduino Autonomous Car – Line Following and Obstacle Stop
User
Controller (Joy-IT R3DIP)
IR Sensors (2 × TCRT5000)
Ultrasonic Sensors (2 × HC-SR04)
Motor Driver & Motors
(L298N, 2 Rear Wheels, 1 Front Rotating Wheel)
Start / Power On Car
Initialize System
«loop» Autonomous Control Loop
Read IR Sensors
Detect Line Position
Read Ultrasonic Sensors
Measure Obstacle Distance
Obstacle within safe distance?
[Yes]
Send Stop Command
Stop Motors
[No]
Line detected?
[No]
Send Stop / Search Command
[Yes]
Line position?
[Left]
[Centered / Both]
[Right]
Command Turn Left
Command Move Forward
Command Turn Right
Stop and Search for Line
Turn Left
Move Forward
Turn Right
[No]
Power off requested?
[Yes]
Power Off / End